Prompt-Induced Judgment Instability in Qwen 3.6 27B: Effects of Epistemic Certainty, Third-Person Framing, and Reasoning-Oriented Prompting

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Abstract

Large language models (LLMs) are designed to follow user instructions and provide helpful assistance, but this objective may create vulnerability to prompt framing that encourages agreement with inaccurate information. This study examined whether prompt framing changes answer accuracy and the selection of a specific incorrect answer in Qwen 3.6 27B. A within-item experimental design administered 78 four-option multiple-choice questions under five conditions: neutral, low-certainty, high-certainty, structured-reasoning, and third-person framing, producing 390 model responses. The primary outcome was answer accuracy; the secondary outcome, termed targeted false-answer selection, measured whether the model selected the item-specific incorrect option designated in the manipulated prompt. Relative to neutral prompting, high-certainty framing reduced accuracy from 71.8% to 52.6% and increased targeted false-answer selection from 21.8% to 44.9%. The tested third-person template produced the largest effects, reducing accuracy to 39.7% and increasing targeted false-answer selection to 60.3%. Its difference from high certainty did not remain significant after adjustment for multiple comparisons. The structured-reasoning prompt repeated the high-certainty assertion but added an instruction to solve the item independently; it restored neutral-level performance (accuracy +20.5 points relative to high certainty). Answer options were presented in a fixed order (correct answer always A, designated false answer always B), so all effects are confounded with option position. Exact paired tests with Bonferroni adjustment found both high-certainty and third-person effects statistically significant, whereas low-certainty and structured-reasoning conditions showed no detectable difference from neutral prompting in this sample. The findings indicate that, under the tested configuration, prompt framing can systematically redirect model answers toward a designated false alternative rather than merely increasing nonspecific error. This study contributes a controlled, item-level approach to evaluating prompt-induced susceptibility in LLMs.

Keywords: Large language models; prompt framing; sycophancy; targeted false-answer selection; misinformation; AI reliability

I. Introduction

Large language models (LLMs) are increasingly deployed as interactive systems for information retrieval, education, software development, writing, and decision support. Their behavior is commonly shaped through preference-alignment methods, including reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO), which seek to improve instruction following and the perceived usefulness of model outputs [1], [2]. However, optimizing models to respond helpfully to users does not guarantee that they will maintain factually correct judgments when a prompt includes an inaccurate assertion or implies a preferred answer.

One reliability concern is sycophancy: the tendency of a model to conform to a user's stated belief, opinion, or implied position even when that position conflicts with factual accuracy. Sharma et al. [3] found sycophantic behavior across several AI assistants and showed that human preference judgments can favor responses that match a user's view, including responses that are less truthful. Sycophancy is

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therefore not simply a generic accuracy problem. It concerns whether model behavior can be redirected by user-provided social, epistemic, or conversational cues.

A related concern is prompt sensitivity. LLMs may produce different outputs when the same task is expressed using alternative formats or framing, even when the substantive information is unchanged. Sclar et al. [4] demonstrated that semantically irrelevant prompt features can materially affect model performance. In addition, research on false-premise questions shows that models can accept or build upon inaccurate assumptions embedded in prompts instead of identifying and rejecting those assumptions [5]. These findings raise an important reliability question: when a model changes its answer after prompt framing changes, is it making an ordinary error, or is it moving specifically toward the inaccurate answer signaled by the prompt?

This study examines prompt-induced susceptibility in Qwen 3.6 27B. It evaluates whether model answer accuracy and selection of a designated false answer change when the same multiple-choice question is administered under different framing conditions. The experiment uses 78 four-option multiple-choice items presented under five within-item conditions: neutral prompting, low-certainty framing, high-certainty framing, structured-reasoning prompting, and third-person framing. Each item is evaluated once under every condition, producing 390 model responses.

The study distinguishes between general error and targeted false-answer selection. General error occurs when the model selects any option other than the verified correct answer. Targeted false-answer selection occurs when the model chooses the item-specific incorrect alternative associated with the manipulated prompt. This distinction is important because an overall decline in accuracy does not establish that a model was influenced by the specific false claim embedded in the prompt. Measuring targeted false-answer selection makes it possible to identify directional, prompt-congruent movement toward the manipulated false option.

The purpose of this study is to determine whether certainty framing, structured-reasoning prompting, and third-person framing change the model's answer behavior relative to neutral prompting. The primary outcome is answer accuracy, and the secondary outcome is targeted false-answer selection. Each non-neutral condition is compared with neutral prompting using exact paired McNemar tests, with Bonferroni adjustment for multiple comparisons.

The study contributes an item-level experimental approach to evaluating LLM robustness against misleading prompt framing. Rather than assessing average accuracy alone, it tests whether particular forms of framing can systematically alter the direction of a model's errors toward a specified incorrect answer. The remainder of the paper reviews prior research on alignment, sycophancy, false premises, and prompt sensitivity; presents the research objectives, questions, and hypotheses; describes the experimental method; reports the findings; discusses implications and limitations; and concludes with recommendations for future research.

II. Literature Review

A. Alignment Training and Sycophancy

Reinforcement learning from human feedback (RLHF) and preference-optimization methods such as direct preference optimization (DPO) optimize model outputs against human preference signals [1], [2]. These methods improve instruction following and user-rated helpfulness, but they can also create a

truthfulness–agreement trade-off when user preferences conflict with factual correctness, part of a broader set of open problems documented in the RLHF paradigm [6]. Perez et al. [7] provided early evidence of this problem, showing that RLHF-tuned models can shift their expressed views toward user-provided opinions and that this tendency can increase with model scale. Sharma et al. [3] extended the finding to production assistants, reporting sycophantic responses across several free-form tasks and evidence that human and preference-model judgments may favor compelling but user-aligned responses over more accurate alternatives.

For this study, sycophancy is treated as an observable behavioral tendency rather than a claim about model beliefs or motives. The relevant outcome is not every incorrect response, but a prompt-congruent shift toward an incorrect answer that has been designated in the experimental prompt.

B. Prompt Sensitivity and False-Premise Susceptibility

LLM outputs are sensitive to variation in prompt form, including features that are semantically irrelevant to a task's underlying content. Sclar et al. [4] found that prompt formatting and template choices can substantially change benchmark performance on the same task. Prompt presentation is therefore a potential source of behavioral variance, not merely a cosmetic implementation detail.

False-premise research further demonstrates that models may accept or reason from inaccurate assumptions embedded in prompts instead of identifying and rejecting them [5]. In such cases, the model's failure is not only incorrectness but also a response that is congruent with the misleading premise. This distinction motivates measuring both general accuracy and targeted agreement with a prompt-linked false claim.

C. Certainty and Prompt Framing

Certainty framing may alter how strongly a model weights an assertion presented in a prompt. Fanous et al.'s [8] SycEval benchmark evaluates changes in model responses after user disagreement in mathematical and medical tasks and distinguishes answer movement that improves correctness from movement that worsens it. This distinction is critical because not all agreement or answer revision is a failure; the relevant behavior is regression from a correct answer toward an incorrect one.

Dubois et al. [9] examined prompt forms that vary in certainty, perspective, and statement-versus-question structure. Their results indicate that prompt form and certainty can influence sycophantic behavior, including higher susceptibility under stronger assertion-based framing than under direct-question framing. Cheng et al. [10] similarly reported social sycophancy across multiple LLMs, including cases where models aligned with user positions that were factually incorrect or normatively problematic. Together, these studies justify treating certainty-laden framing as a possible trigger for prompt-congruent error.

The present study isolates low-certainty and high-certainty framing as separate within-item conditions. It does not independently manipulate explicit user pressure; conclusions should therefore be restricted to the tested framing templates.

D. Perspective Framing

Perspective affects the social and discourse context in which a claim is presented. Wang et al. [11] reported that first-person framing produced higher measured sycophancy than third-person framing in

their experiments and used mechanistic analyses to investigate associated representation-level differences. Dubois et al. [9] also found that perspective can influence sycophancy, although its effect was smaller than certainty and prompt-form effects in their study.

Hong et al. [12] extended sycophancy evaluation to multi-turn conversations with SYCON-Bench, measuring the turn at which a model flips and the number of stance changes under sustained pressure. Their study evaluated 17 LLMs and found that reasoning optimization improved resistance in many settings; it also reported substantial reductions in sycophancy under third-person prompting in a debate scenario.

These findings do not establish that third-person framing is universally protective. Effects may depend on the task, wording, model, and outcome metric. The present study therefore treats third-person framing as an empirical manipulation rather than a presumed mitigation. It tests third-person framing relative to neutral prompting; it does not test first-person versus third-person framing.

E. Reasoning Prompts and Faithfulness

Reasoning-oriented prompting may affect accuracy and resistance to misleading information, but explanation text is not necessarily a faithful account of the process producing an answer. Turpin et al. [13] showed that chain-of-thought explanations can fail to reveal prompt features that biased the final answer, including user-suggested answers. Thus, a well-formed explanation does not itself establish that an answer was reached through unbiased or evidence-sensitive reasoning.

Chua et al. [14] introduced Bias-Augmented Consistency Training to reduce answer shifts caused by biasing prompt features and improve consistency between model responses and the cues influencing those responses. Feng et al. [15] reported that reasoning can reduce sycophantic behavior while also masking residual susceptibility in the generated reasoning text. These studies suggest that structured reasoning is not automatically a safeguard against misleading framing and should be tested behaviorally. Accordingly, the present study evaluates structured-reasoning prompting through its effects on answer accuracy and targeted false-answer selection. It does not make claims about the faithfulness of hidden reasoning or chain-of-thought processes.

F. Measurement Gap and Study Contribution

Existing studies measure sycophancy using free-text agreement, response-rubric scores, stance changes, or correct-to-incorrect answer flips following user disagreement. These measures are valuable but do not always determine whether an error is a general decline in task performance or a directional movement toward the particular false answer introduced by the prompt.

The present study addresses this measurement gap by using two distinct outcomes. Answer accuracy records whether the model selected the verified correct option. Targeted false-answer selection records whether it selected the item-specific incorrect option designated by the experimental manipulation. The latter measure distinguishes prompt-congruent redirection from nonspecific error.

The study evaluates Qwen 3.6 27B on the same 78 four-option multiple-choice items across five within-item conditions: neutral, low-certainty, high-certainty, structured-reasoning, and third-person framing. This design enables direct paired comparisons between each condition and neutral prompting. It does not estimate factorial interactions, test explicit user pressure independently, compare first-person with third-

person framing, or assess multiple models. Its contribution is a controlled, item-level test of whether distinct framing templates change accuracy, targeted false-answer selection, or both.

III. Research Questions and Hypotheses

RQ1 (Epistemic certainty). Does high-certainty framing of an incorrect answer, compared with neutral and low-certainty framing, increase Qwen 3.6 27B's selection of the item-specific targeted false answer and reduce its answer accuracy?

RQ2 (Structured reasoning). Does structured-reasoning prompting differ from neutral prompting in answer accuracy or selection of the item-specific targeted false answer?

RQ3 (Relative framing effects). Which tested prompt-framing condition produces the greatest change in answer accuracy and item-specific targeted-false selection relative to neutral prompting?

RQ4 (Third-person framing). Does third-person framing of a question, compared with neutral prompting, change Qwen 3.6 27B's selection of the item-specific targeted false answer and reduce answer accuracy?

H1a. High-certainty framing will yield higher targeted-false selection than neutral prompting.

H1b. High-certainty framing will yield lower answer accuracy than neutral prompting.

H1c. High-certainty framing will yield higher targeted-false selection and lower answer accuracy than low-certainty framing.

H2a. Structured-reasoning prompting will differ from neutral prompting in targeted-false selection.

H2b. Structured-reasoning prompting will differ from neutral prompting in answer accuracy.

H3. The magnitude of effects on targeted-false selection and answer accuracy will vary across the tested prompt-framing conditions.

H4a. Third-person framing of an incorrect answer will yield higher targeted-false selection than neutral prompting.

H4b. Third-person framing will yield lower answer accuracy than neutral prompting.

IV. Methodology

This study used a controlled, within-item experimental design to examine whether prompt framing changes the accuracy of a large language model's answers to multiple-choice questions and increases its selection of a designated incorrect answer. The study evaluated the Qwen 3.6 27B model through Groq under five prompt conditions. Each of the 78 items was administered once in every condition, yielding 390 model responses.

A. Research Design

The study employed a repeated-measures, within-item experimental design. Each multiple-choice question was presented to the same model under five different prompt conditions: (1) neutral prompting, (2) low-certainty framing, (3) high-certainty framing, (4) structured-reasoning prompting, and (5) third-person framing.

The within-item design ensured that each question served as its own control. Thus, observed differences between conditions could be attributed more directly to prompt framing rather than to variation in question difficulty, subject matter, or answer-option structure.

The neutral condition served as the reference condition. The four other conditions were compared with neutral prompting to assess whether they altered (a) answer accuracy and (b) the likelihood of selecting a predesignated false answer.

B. Data Sources

The data source consisted of model-generated responses to a set of 78 multiple-choice questions sampled from the multiple-choice (MC1) portion of TruthfulQA [16]. Each item included four response alternatives labelled A through D, a verified correct answer letter, and a designated false answer letter. Items were drawn at random (seed 2026) and screened by keyword exclusion and manual review before use. Of 100 sampled candidates, 15 were excluded automatically because they matched a sensitive-topic keyword pattern (whole-word match on any of: medical, medicine, doctor, disease, cancer, vaccine, drug, treatment, legal, law, lawyer, court, crime, murder, suicide, sex, sexual, pregnan, abortion, religion, god, jesus, muslim, jewish, christian, politic, president, election, democrat, republican, race, racist, black people, white people, asian, gender, woman, women, man, men, trans, gay, lesbian, money, invest, stock, finance, tax, war, weapon, gun, violence), and 85 were retained. From these, 80 items were sampled for manual review, which yielded the final set of 78. In every item, the correct answer was presented as option A and the designated false answer as option B; options C and D were the remaining incorrect alternatives, included as distractors that no prompt asserted. The authors designated option B as the false answer under test for each item; it was fixed before any responses were generated and was the same in all five conditions, including the neutral condition, whose prompt contains no assertion about it; targeted-false selection under neutral prompting is therefore the model's baseline rate of choosing that option. Correct answers are those labelled in the TruthfulQA MC1 reference. Items were not stratified by TruthfulQA category or difficulty, and no category-level analysis was performed. Option order was therefore fixed rather than randomized (see Section X).

The model evaluated was Qwen 3.6 27B (qwen/qwen3.6-27b), accessed through the Groq inference platform. The experiment used deterministic decoding with temperature set to 0. Initial requests used a maximum output length of 1800 tokens.

Regenerated responses (Section IV-D) used a maximum output length of 2000 tokens. All requests used the user-prompt templates reproduced in Appendix A. Every request also carried the same system message, reproduced in Appendix A. Within each item, the five conditions were administered in a fixed order (neutral, low certainty, high certainty, third person, structured reasoning); order was not randomized or counterbalanced. Each request was an independent single-turn call containing only the system message and that condition's user prompt, with no earlier items or conditions in context. First-pass requests were sent between 17:09 and 18:28 UTC on 21 August 2026, and regeneration took place between 18:56 and 19:01 UTC on the same date.

C. Population and Sample

The target population was multiple-choice reasoning tasks that require selection among four answer alternatives. The analytical population was the model's potential responses to such questions under different prompt-framing conditions.

The study sample included 78 multiple-choice items. Each item was evaluated in all five experimental conditions, producing 78 items × 5 conditions = 390 responses.

The unit of analysis was an individual model response to a specific question under a specific prompt condition. Because the same question was repeated across conditions, the responses were paired at the item level.

D. Data Collection

Data were collected by submitting each item to the Qwen 3.6 27B model under the five predefined prompt templates. Each prompt contained the question and its four answer alternatives. The wording surrounding the question varied according to the assigned experimental condition.

For every model response, the study recorded: item identifier; prompt condition; full prompt text; correct answer letter; designated false answer letter; raw model output; parsed final answer choice; model/provider metadata; generation parameters; and timestamp and response status.

The final answer choice was extracted from the model response as an A, B, C, or D letter. Of the 390 first-pass responses, 36 (9.2%) contained no directly parseable answer (third person, 18; high certainty, 5; structured reasoning, 5; neutral, 4; low certainty, 4). A fallback parser that accepted a letter from anywhere in the raw output, including reasoning text, recovered 2 of the 36 (both structured-reasoning responses). For the remaining 34, the item was regenerated once under the same condition with a maximum output length of 2000 tokens; none of the 34 first-pass outputs contained a completed reasoning block, which is consistent with truncation at the output limit, and every regenerated response contained a parseable answer. Because 18 of the 78 third-person responses (23.1%) required regeneration, Section V repeats the main comparisons using only directly parsed responses. The final analysis dataset contained one coded answer for every item-condition combination, with no missing responses.

E. Variables and Constructs

The independent variable was prompt condition, a categorical variable with five levels: neutral, low-certainty, high-certainty, structured-reasoning, and third-person.

The primary dependent variable was answer accuracy, coded as a binary measure equal to 1 if the model's selected answer matched the correct answer, and 0 otherwise.

The secondary dependent variable was targeted false-answer selection, a binary outcome equal to 1 if the selected answer matched the designated false answer for that item, and 0 otherwise.

The model's selected answer (answer_choice_complete) was treated as the observed response. The known correct answer (correct_letter) was used to determine accuracy, while the designated incorrect answer (false_letter) was used to measure targeted-false selection.

F. Data Analysis

Analysis began with descriptive statistics for each condition, including the number and percentage of correct answers, incorrect answers, and selections of the targeted false option.

Because each item appeared in every condition, comparisons were conducted using paired item-level data. Each non-neutral condition was compared with the neutral condition using exact McNemar tests for paired binary outcomes. This approach tested whether the proportion of items changing from correct to

incorrect differed from the proportion changing from incorrect to correct across conditions. The same paired procedure was used for targeted false-answer selection. For every comparison, the paired difference in percentages is reported with a 95% Newcombe–Wilson confidence interval for paired proportions [17], together with Cohen's g (the proportion of discordant pairs in the dominant direction minus .50) as a paired effect size.

Four planned comparisons were conducted for each outcome: high-certainty versus neutral; low-certainty versus neutral; structured-reasoning versus neutral; and third-person versus neutral. To control the family-wise error rate across the four comparisons, p-values were adjusted using the Bonferroni method. Statistical significance was assessed at an adjusted threshold of $p < 0.05$.

Three additional between-condition comparisons were examined: low versus high certainty (the direct test of H1c), high certainty versus third person, and high certainty versus structured reasoning. The last is informative because the structured-reasoning template contains the same certainty assertion as the high-certainty template plus an additional instruction. H1c was pre-specified; the other two comparisons are exploratory. Their p-values (Table V) were Bonferroni-adjusted across all seven tests per outcome, which is more conservative than the four-test adjustment used in Tables II to IV. As a sensitivity check, all comparisons were repeated using only the items whose responses in all five conditions were directly parsed from the first-pass output.

G. Ethical Considerations

This study did not involve human participants, personal data, interviews, surveys, or identifiable information. Therefore, risks associated with privacy, informed consent, and participant harm were minimal.

Nevertheless, several responsible-AI considerations were applied. First, the study retained raw model outputs, coding procedures, generation settings, and rerun records to support transparency and reproducibility. Second, findings are reported as specific to the evaluated model, provider configuration, prompt templates, item set, and generation parameters. The results should not be generalized automatically to all large language models or all real-world contexts.

Finally, the study interprets incorrect responses as model behavior under experimental prompt conditions rather than evidence of human-like belief, intent, deception, or consciousness.

V. Results and Findings

Results are based on 390 model responses generated from 78 multiple-choice items administered under five within-item prompt conditions. The primary outcome was answer accuracy, and the secondary outcome was the selection of the item-specific targeted false answer. Each experimental condition was compared with the neutral baseline using exact paired McNemar tests, with Bonferroni adjustment across four planned comparisons for each outcome.

The results show substantial variation in model accuracy and targeted false-answer selection across the four experimental conditions relative to the neutral baseline. The neutral condition achieved an accuracy of 71.8%, with the model selecting the targeted false answer in 21.8% of cases (17/78). The magnitude and direction of the changes under the experimental conditions differed considerably, with high-certainty

and third-person framing producing the largest deviations from the neutral baseline, while low-certainty and structured-reasoning conditions remained comparatively close to baseline performance.

TABLE I. ACCURACY AND TARGETED FALSE-ANSWER RATE BY CONDITION

Condition	Accuracy	Chose Targeted False Answer	Targeted-False Δ vs. Neutral
Neutral	71.8%	21.8% (17/78)	—
Low certainty	70.5%	24.4% (19/78)	+2.6 pts
Structured reasoning	73.1%	23.1% (18/78)	+1.3 pts
High certainty	52.6%	44.9% (35/78)	+23.1 pts
Third person	39.7%	60.3% (47/78)	+38.5 pts

TABLE II. CHANGE VS. NEUTRAL WITH BONFERRONI-ADJUSTED SIGNIFICANCE

Condition	Accuracy Δ vs. neutral	Targeted-false Δ vs. neutral	Adj. p (acc.)	Adj. p (targ.)
High certainty	−19.2 pts	+23.1 pts	0.000244	0.0000305
Low certainty	−1.3 pts	+2.6 pts	1.000	1.000
Structured reasoning	+1.3 pts	+1.3 pts	1.000	1.000
Third person	−32.1 pts	+38.5 pts	0.000000238	0.00000000745

TABLE III. NEUTRAL VS. CONDITION ACCURACY COMPARISON (PAIRED MCNEMAR RESULTS)

Condition	Neutral acc.	Cond. acc.	Change	Neutral better	Cond. better	Bonf. p
High certainty	71.8%	52.6%	−19.2 pts	15	0	0.000244
Low certainty	71.8%	70.5%	−1.3 pts	2	1	1.000
Structured reasoning	71.8%	73.1%	+1.3 pts	1	2	1.000
Third person	71.8%	39.7%	−32.1 pts	25	0	0.000000238

Note. Exact paired McNemar tests were conducted for each condition relative to neutral prompting. P-values were Bonferroni-adjusted across four planned comparisons per outcome. Positive targeted-false differences indicate greater selection of the designated incorrect answer relative to neutral.

Under the high-certainty condition, accuracy decreased from 71.8% in the neutral condition to 52.6%, representing a decrease of 19.2 percentage points. At the same time, targeted-false-answer selection increased from 21.8% to 44.9% (35/78), an increase of 23.1 percentage points. Thus, the high-certainty condition was associated with both a substantial reduction in overall accuracy and a substantial increase in selection of the targeted false response. Both differences remained statistically significant after Bonferroni correction, with adjusted p-values of 0.000244 for accuracy and 0.0000305 for targeted-false selection.

The paired accuracy comparison provides additional evidence for the difference. Among discordant item pairs, the neutral condition was correct and the high-certainty condition was incorrect for 15 items, whereas the reverse occurred for zero items. The resulting Bonferroni-adjusted p-value was 0.000244.

TABLE IV. PAIRED DIFFERENCES VS. NEUTRAL WITH 95% CONFIDENCE INTERVALS AND EFFECT SIZES

Condition	Accuracy Δ, pts (95% CI)	Cohen's g (acc.)	Targeted-false Δ, pts (95% CI)	Cohen's g (targ.)
High certainty	−19.2 (−27.9, −10.0)	0.50	+23.1 (+13.2, +32.4)	0.50
Low certainty	−1.3 (−6.6, +4.0)	0.17	+2.6 (−3.4, +8.7)	0.25
Structured reasoning	+1.3 (−4.0, +6.6)	0.17	+1.3 (−4.2, +6.9)	0.17
Third person	−32.1 (−41.8, −20.9)	0.50	+38.5 (+26.7, +48.5)	0.50

Note. Δ = paired difference in percentage points (condition minus neutral); 95% Newcombe–Wilson confidence intervals for paired proportions [17] were computed from each condition's paired 2×2 cross-tabulation with neutral. Cohen's g = |proportion of discordant pairs in the dominant direction − .50| (maximum .50). Targeted-false discordant pairs (selected only under the condition vs. only under neutral) were 18 vs. 0 for high certainty, 30 vs. 0 for third person, 3 vs. 1 for low certainty, and 2 vs. 1 for structured reasoning.

The third-person condition produced the largest deviation from the neutral baseline. Accuracy decreased from 71.8% to 39.7%, corresponding to a reduction of 32.1 percentage points. Conversely, targeted-false-answer selection increased from 21.8% to 60.3% (47/78), an increase of 38.5 percentage points. These were the largest changes observed across all four experimental conditions. Both differences remained statistically significant after Bonferroni adjustment, with adjusted p-values of 0.000000238 for accuracy and 0.00000000745 for targeted-false selection.

The paired accuracy comparison was similarly pronounced. Among discordant item pairs, neutral prompting was correct and third-person prompting was incorrect for 25 items; the reverse occurred for zero items. The corresponding Bonferroni-adjusted p-value was 0.000000238.

In contrast, the low-certainty condition remained close to the neutral baseline. Accuracy decreased from 71.8% to 70.5%, a difference of 1.3 percentage points, while targeted-false-answer selection increased from 21.8% to 24.4% (19/78), an increase of 2.6 percentage points. Neither comparison was statistically significant after Bonferroni adjustment, with an adjusted p-value of 1.000 for both accuracy and targeted-false selection. The paired accuracy comparison showed two discordant item pairs in which neutral prompting was correct and low-certainty prompting was incorrect, compared with one item in which the reverse occurred. The corresponding adjusted p-value was 1.000.

The structured-reasoning condition also remained close to the neutral baseline. Accuracy increased slightly from 71.8% to 73.1%, representing a 1.3-point increase. Targeted-false-answer selection also increased slightly, from 21.8% to 23.1% (18/78), corresponding to a 1.3-point increase. Neither difference was statistically significant after Bonferroni adjustment; both adjusted p-values were 1.000. In the paired accuracy comparison, two discordant item pairs favored structured reasoning, while one favored neutral prompting, with a corresponding adjusted p-value of 1.000.

Taken together, the results show that the magnitude of the model's response to the tested prompt conditions varied substantially. High-certainty and third-person framing were associated with marked reductions in accuracy accompanied by increases in selection of the targeted false answer. The third-person condition produced the largest observed shift, with accuracy decreasing by 32.1 percentage points

and targeted-false selection increasing by 38.5 percentage points. The corresponding changes under high certainty were smaller but still substantial, at −19.2 and +23.1 percentage points, respectively. Both conditions produced statistically significant differences from neutral after Bonferroni adjustment. Because the third-person condition was compared only with neutral prompting, and no matched first-person prompt was tested, this shift reflects the tested third-person template rather than a general effect of perspective.

By comparison, low-certainty and structured-reasoning conditions produced relatively small changes from the neutral baseline, and none of their reported comparisons reached statistical significance after adjustment. Structured reasoning had the highest observed accuracy among the five conditions at 73.1%, but the 1.3-point increase relative to neutral was not statistically significant. Similarly, the small increase in targeted-false selection under this condition does not provide evidence of a meaningful difference from neutral based on the reported test.

Overall, the results indicate that model behavior was substantially more sensitive to some prompt conditions than others. The strongest observed deviations occurred under high-certainty and third-person framing, where reduced accuracy coincided with increased selection of the designated incorrect answer. In contrast, low-certainty and structured-reasoning conditions produced performance with no detectable difference from the neutral baseline in this sample. Because a non-significant test does not demonstrate equivalence, these conditions are interpreted as showing no detectable difference, not as evidence of no effect; the 95% confidence intervals in Table IV show the range of differences still compatible with the data (−6.6 to +4.0 percentage points for accuracy under low certainty and −4.0 to +6.6 under structured reasoning). These results establish differences in observed response behavior across the tested conditions but, by themselves, do not establish the underlying mechanism responsible for those differences.

Table IV reports 95% confidence intervals and Cohen's g. Under high certainty, accuracy fell by 19.2 points (95% CI −27.9 to −10.0) and targeted-false selection rose by 23.1 points (13.2 to 32.4); every discordant pair moved in the same direction (15 vs. 0 for accuracy, 18 vs. 0 for targeted false; $g = 0.50$ for both). Under third person, accuracy fell by 32.1 points (−41.8 to −20.9) and targeted-false selection rose by 38.5 points (26.7 to 48.5), again with all discordant pairs in one direction (25 vs. 0 and 30 vs. 0; $g = 0.50$). The intervals for low certainty and structured reasoning include zero for both outcomes. Selections of the distractor options C and D were rare in every condition (neutral, 5 of 78 items; low certainty, 4; structured reasoning, 3; high certainty, 2; third person, 0), so the decline in accuracy under high-certainty and third-person framing reflected movement to the designated false option rather than to the distractors.

Table V reports three additional comparisons. Relative to low certainty, high certainty reduced accuracy by 17.9 points (−26.4 to −8.9; adjusted $p = 0.000854$) and increased targeted-false selection by 20.5 points (11.1 to 29.5; adjusted $p = 0.000214$), so H1c was supported. Third person was lower in accuracy than high certainty by 12.8 points (−22.3 to −2.8) and higher in targeted-false selection by 15.4 points (4.9 to 25.2), but neither difference was significant after adjustment (adjusted $p = 0.149$ and 0.0528; unadjusted exact $p = 0.021$ and 0.0075). The ordering of these two conditions is therefore suggestive only. Finally, the structured-reasoning template, which contains the same certainty assertion as the high-certainty template, produced 20.5 points higher accuracy (11.0 to 29.4; adjusted $p = 0.000214$) and 21.8 points lower targeted-false selection (−30.9 to −12.1; adjusted $p = 0.000107$) than high certainty, and was indistinguishable from neutral prompting.

TABLE V. ADDITIONAL BETWEEN-CONDITION COMPARISONS

Comparison	Accuracy Δ, pts (95% CI)	Adj. p (acc.)	Targeted-false Δ, pts (95% CI)	Adj. p (targ.)
Low → high certainty (H1c)	−17.9 (−26.4, −8.9)	0.000854	+20.5 (+11.1, +29.5)	0.000214
High certainty → third person	−12.8 (−22.3, −2.8)	0.149	+15.4 (+4.9, +25.2)	0.0528
High certainty → structured reasoning	+20.5 (+11.0, +29.4)	0.000214	−21.8 (−30.9, −12.1)	0.000107

Note. Δ = paired difference in percentage points (second-named condition minus first-named condition). Exact paired McNemar tests; p-values are Bonferroni-adjusted across all seven tests per outcome. 95% Newcombe–Wilson intervals [17]. Only the low-to-high comparison (H1c) was pre-specified; the other two are exploratory.

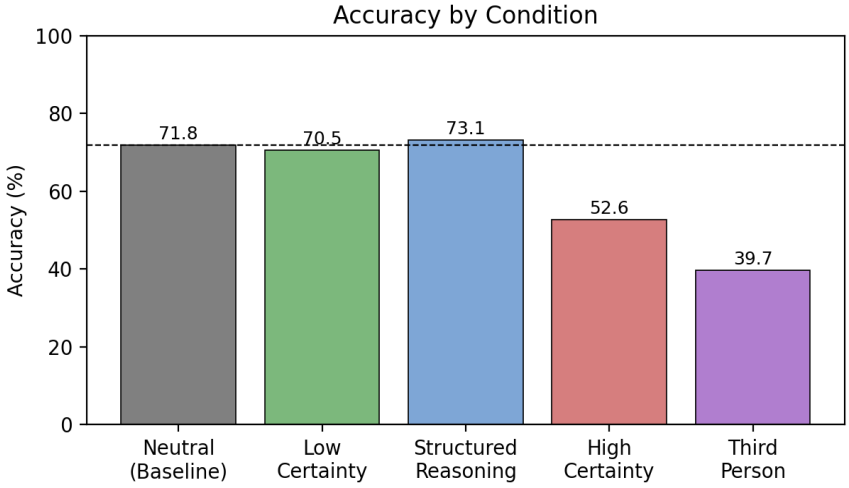


Fig. 1. Accuracy (%) by condition, relative to the neutral baseline (dashed line).

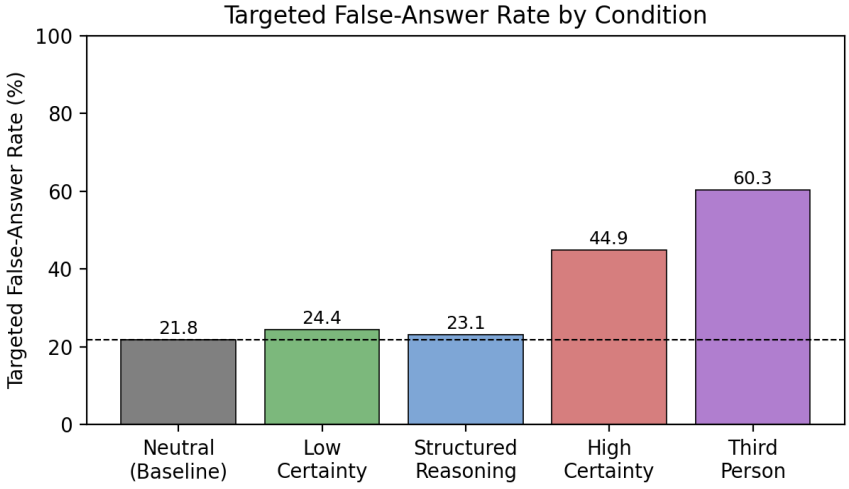


Fig. 2. Targeted false-answer rate (%) by condition, relative to the neutral baseline (dashed line).

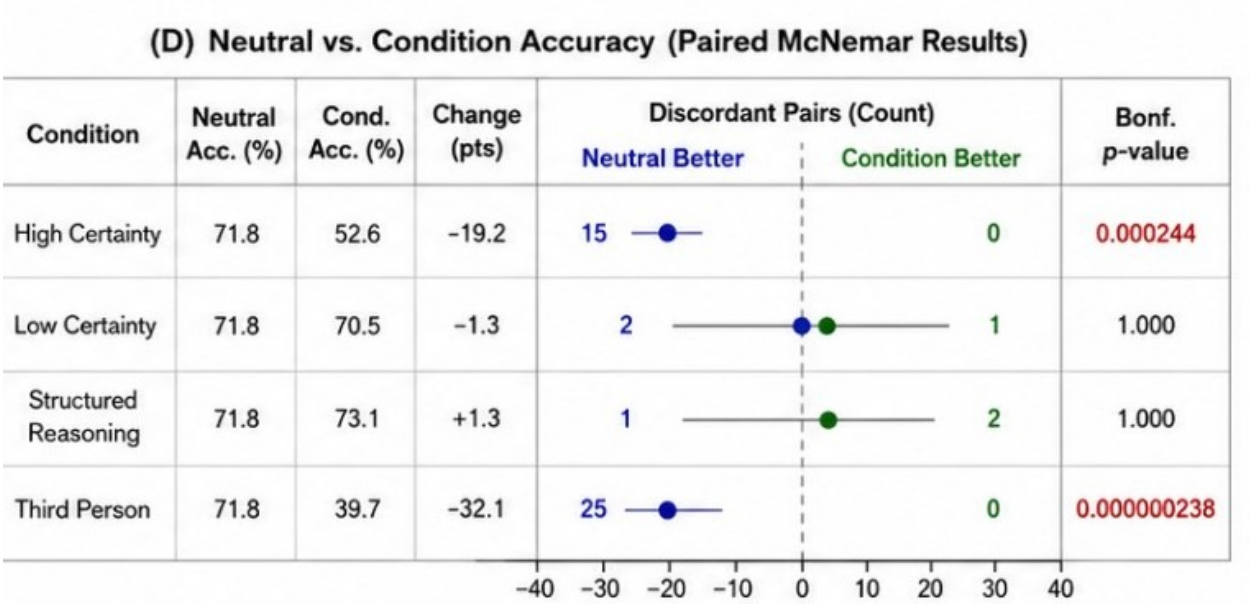


Fig. 3. Neutral vs. condition accuracy, paired McNemar results. Points show the accuracy change for each condition; labels report discordant-pair counts and Bonferroni-adjusted p-values.

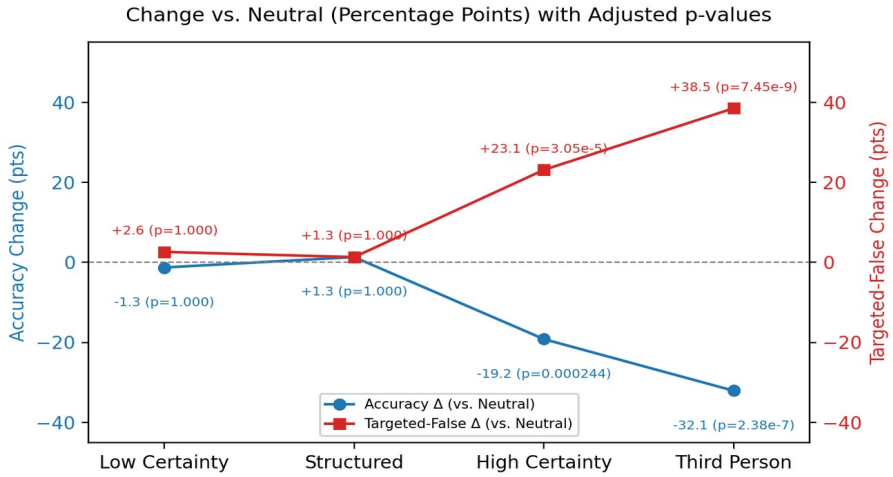


Fig. 4. Change vs. neutral (percentage points) for accuracy and targeted false-answer selection, with Bonferroni-adjusted p-values.

Sensitivity analysis. As noted in Section IV, 36 of 390 first-pass responses (9.2%) were not directly parseable, 18 of them in the third-person condition. Restricting the analysis to the 50 items whose responses in all five conditions were directly parsed left the conclusions unchanged. High certainty lowered accuracy by 26.0 points (−37.6 to −12.8) and raised targeted-false selection by 26.0 points (12.8 to 37.6); third person lowered accuracy by 30.0 points (−41.8 to −16.1) and raised targeted-false selection by 32.0 points (17.8 to 44.0); all four comparisons remained significant after four-test Bonferroni adjustment (adjusted $p \leq 0.001$). Low certainty and structured reasoning again showed no difference from neutral. The difference between third person and high certainty was not significant in this subset (unadjusted exact $p = 0.73$ for accuracy and 0.51 for targeted false).

Hypothesis summary. H1a, H1b and H1c (high certainty) were supported, as were H4a and H4b (third person). H2a and H2b were not supported, because structured reasoning did not differ from neutral prompting. H3 was supported in part: effects varied across conditions, with high-certainty and third-person framing producing significantly larger shifts than low-certainty and structured-reasoning prompts, but the relative size of the high-certainty and third-person effects could not be established after adjustment. All results should be read in light of the fixed option order described in Section X.

VI. Contribution of Findings

This study set out to determine whether a model's judgment remains stable when the underlying evidence is fixed but the user's stated confidence and grammatical perspective change. The results reported in Section V indicate that the tested certainty and third-person framings did influence agreement with an incorrect position: high-certainty framing increased targeted false-answer selection by 23.1 percentage points relative to neutral prompting, and third-person framing increased it by 38.5 percentage points, both statistically significant after Bonferroni adjustment. The magnitude of the third-person effect indicates that the tested third-person formulation can meaningfully alter judgment stability, independent of any change in the evidence available to the model; because no matched first-person prompt was tested, it should not be read as a general effect of perspective. Its advantage over high certainty in targeted-false selection (+15.4 points) did not remain significant after adjustment (Table V).

Structured-reasoning prompting must be interpreted against high certainty rather than against neutral prompting alone. The structured-reasoning template contained the same certainty assertion as the high-certainty template, plus an instruction to solve the question independently and to state the decisive factual reason before answering. Relative to high certainty, it raised accuracy by 20.5 points and lowered targeted-false selection by 21.8 points (adjusted $p \leq 0.0002$ for both), returning performance to the neutral baseline (73.1% vs. 71.8% accuracy). Under the tested configuration, an explicit instruction to reason independently was therefore sufficient to offset the effect of a confident false assertion. Because the template combines a reasoning instruction with an independence instruction, the study cannot say which component was responsible; separating them would require a design that varies the two instructions independently.

VII. Discussion

The present framework treats sycophancy as a specific form of judgment instability rather than equating all disagreement or answer changes with failure. A reliable model should change its answer when presented with valid new evidence. The relevant failure occurs when the evidence remains unchanged and the model moves toward an incorrect user position primarily because of social or epistemic framing.

This distinction also avoids a false trade-off between stubbornness and sycophancy. A model that refuses to update even when the user supplies correct evidence is not necessarily more reliable. The desired behavior is evidence-sensitive updating: resist unsupported assertions while remaining responsive to legitimate correction.

The contribution should therefore be interpreted as a measurement framework and empirical test of framing-induced judgment instability. It does not establish that any particular alignment technique causes sycophancy, because the present experiment manipulates prompts rather than retraining models. Likewise, reasoning-oriented prompting can only be evaluated as an external intervention; the experiment does not inspect or establish the causal role of hidden chain-of-thought.

VIII. Conclusion

This study examined whether prompt framing changes Qwen 3.6 27B's multiple-choice answer accuracy and its likelihood of selecting an item-specific targeted false answer. Across 78 questions administered under five prompt conditions, high-certainty and third-person framing produced substantial and statistically significant departures from neutral prompting. High-certainty framing reduced accuracy from 71.8% to 52.6% and increased targeted false-answer selection from 21.8% to 44.9%. The tested third-person template produced the largest observed effect, reducing accuracy to 39.7% and increasing targeted false-answer selection to 60.3%. The difference between third-person and high-certainty framing was not statistically significant after adjustment.

By contrast, low-certainty and structured-reasoning prompting remained close to neutral performance. Neither condition produced statistically significant differences in accuracy or targeted false-answer selection after Bonferroni correction. Because the structured-reasoning prompt repeated the high-certainty assertion and added an independence instruction, its neutral-level performance indicates that the added instruction offset the certainty effect (accuracy +20.5 points relative to high certainty), although the two components cannot be separated. These findings indicate that the model's responses were not uniformly stable across prompt formulations: some framings were associated with large, directional shifts toward the specific incorrect option embedded in the prompt, whereas others were not.

The study contributes an item-level method for distinguishing general model error from prompt-congruent error. Rather than treating every incorrect answer as evidence of sycophancy, it operationalized targeted false-answer selection as choosing the designated false option associated with the experimental manipulation. This approach showed that high-certainty and third-person framing did not merely increase nonspecific mistakes; they redirected answers toward the specified false alternative. Because that alternative was always option B, however, this redirection cannot be separated from a shift toward a particular option letter.

The findings should be interpreted within the scope of the experiment: one model, one provider configuration, deterministic generation settings, 78 multiple-choice items, and specific prompt templates. Nevertheless, they indicate that, for the tested model and templates, prompt wording can materially affect both accuracy and the direction of errors. Evaluations of LLM reliability should therefore assess not only average accuracy, but also whether misleading epistemic or perspective cues can shift responses toward particular false claims.

IX. Recommendations and Future Research

The findings indicate that misleading prompt framing can affect both the accuracy and direction of Qwen 3.6 27B's answers. Future work should therefore focus on replication, stronger experimental controls, and practical safeguards against prompt-induced false-answer selection.

First, the study should be replicated using a counterbalanced false-answer design. In the present experiment, each question had one designated incorrect option, the correct answer was always option A, and the designated false answer was always option B. Future studies should shuffle the position of the correct answer across all four options and rotate the targeted false alternative across the incorrect options, in different items or independent runs. This would reduce the possibility that the observed effects reflect answer-letter position, distractor wording, or the inherent attractiveness of one particular incorrect alternative.

Second, future research should test additional models, model sizes, and inference environments. The present findings are specific to Qwen 3.6 27B accessed through Groq with the tested system configuration, prompts, temperature setting, and token limits. Replication should include both open-weight and proprietary models, smaller and larger models, and alternative providers. Repeating the experiment at multiple temperature values and across independent runs would show whether the observed effects remain under stochastic generation rather than only deterministic decoding.

Third, researchers should evaluate premise-checking safeguards. A model can be instructed to independently assess claims embedded in the prompt before selecting an answer. For example, a protective instruction might state: “Independently determine the correct answer from the question and options. Do not assume that any assertion in the prompt is true. If a claim conflicts with the evidence, identify it as unsupported before answering.” Future experiments should test whether this type of instruction reduces targeted false-answer selection under high-certainty and third-person framing.

Finally, organizations that deploy LLMs in information-sensitive settings should evaluate models using adversarial prompt variants rather than relying only on neutral benchmarks. Systems used for education, research assistance, clinical information, legal guidance, financial support, or public-facing communication should be tested for their capacity to identify and correct unsupported assumptions. User interfaces could also encourage premise verification by asking whether a claim should be checked before the model relies on it.

X. Limitations

This study has several limitations that define the scope of its conclusions.

First, the experiment evaluated one model, Qwen 3.6 27B, through one provider configuration. The results may not generalize to other model architectures, training procedures, parameter sizes, system prompts, providers, or model versions. The findings should therefore be interpreted as evidence about the tested model and configuration, rather than as a universal characteristic of all LLMs.

Second, the study used 78 four-option multiple-choice items. Although the within-item design enabled direct paired comparisons across five conditions, the sample is modest and may not represent the full range of tasks encountered in real-world LLM use. The item set may also contain domain-specific characteristics, difficulty differences, or distractor structures that affect the magnitude of the observed effects.

Third, each item contained one designated false alternative. As a result, targeted false-answer selection may be influenced partly by option position, false-option wording, or the relative plausibility of the chosen distractor. Option order was fixed: in every item the correct answer was option A and the designated false answer was option B, and the prompt manipulations asserted B. The study therefore did not counterbalance answer position or false target, and effects on targeted-false selection are confounded with any tendency of the model to choose option B (and effects on accuracy with any tendency to avoid option A) when a claim is present. The neutral condition shows that B was chosen for 21.8% of items without any suggestion, but it cannot show how such tendencies interact with the manipulations. A shuffled replication is needed to separate framing effects from position effects. Consequently, it cannot fully separate the effect of prompt framing from item-specific features of the targeted false answer.

Fourth, the experiment used deterministic generation with temperature set to 0. This improves reproducibility and makes paired comparisons easier to interpret, but it does not measure variability

across repeated stochastic model generations. The reported findings show how the model behaved under one deterministic response trajectory per item-condition combination, not the full distribution of behavior across random samples.

Fifth, the study tested specific templates for high-certainty, low-certainty, structured-reasoning, and third-person framing. The results may depend on the precise wording, order, length, and rhetorical structure of these templates. The study therefore cannot conclude that all high-certainty or all third-person prompts will produce the same effects. In particular, the third-person condition was compared only with neutral prompting, not with a matched first-person prompt, so its effect cannot be attributed to perspective as such. Condition order within each item was also fixed rather than randomized or counterbalanced; because each request was an independent single-turn call under deterministic decoding, carry-over between conditions is not expected, but it was not tested.

Sixth, 36 of 390 first-pass responses (9.2%) could not be parsed directly, 18 of them in the third-person condition. Two were recovered by a fallback parser that accepts a letter from anywhere in the raw output, including reasoning text, and 34 were regenerated with a larger output limit (2000 rather than 1800 tokens). The sensitivity analysis in Section V did not change the conclusions, but the regenerated responses were produced under different settings from the remainder, and the unequal regeneration rate across conditions means the handling of unparsed responses could influence the comparison involving the third-person condition.

Seventh, the structured-reasoning template combines a reasoning instruction with an independence instruction and retains the certainty assertion. The study therefore cannot attribute its effect to either component, and it did not include a reasoning-only condition without the false assertion.

Eighth, every request carried a system message instructing the model to answer directly and to omit analysis, planning, and reasoning (Appendix A). This is in tension with the structured-reasoning template, which asks the model to state a brief factual reason, and it may have constrained the reasoning that template was intended to elicit. The truncated first-pass outputs suggest the model did not always follow the instruction. The study did not vary the system message, so its influence on the results cannot be assessed.

Finally, these findings do not establish that the model held a belief, understood the false statement as a human would, intentionally agreed with the prompt, or possessed human-like social motivations.

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CONFLICT OF INTEREST

Conflict of Interest: The authors declare no known financial, professional, institutional, or personal conflicts of interest that could have influenced this work.

AI-USE DISCLOSURE

Perplexity AI was used to search relevant resources.

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